Progress Report — ACCESS project AGR260032

Leakage-controlled, evidence-aware gene annotation for plant genomes. PI: Won Yim, University of Nevada, Reno. Resource: NCSA DeltaAI (GH200). Report date: October 3, 2026.

1. Project goal

We are training a deep-learning gene-structure annotation model for plant genomes. A model of about 400 million parameters reads up to 129,024 nt of genomic sequence and outputs gene structures. A sequence-only baseline is compared with a new model that also reads evidence tracks from ESTs, PacBio and ONT long reads, and cross-species protein alignments. Every evidence record is filtered so that no held-out locus used for validation or testing can leak into training inputs. The pre-registered question is whether the evidence improves gene prediction beyond the sequence-only model, tested with control experiments that isolate what part of the evidence drives any gain.

2. Use of the current allocation

Of 1,499 GPU-hours awarded, about 650 have been used and 849 remain.

Activity	GPU-h	Outcome
Sequence-only baseline, seed 123, 4 GPUs, NGC PyTorch 26.08 with torch.compile	630	Epoch 13; validation loss still improving; early stopping running
Runtime qualification over three sequence-length tiers, 1–4 GPUs	13	Production runtime selected; time and memory measured per tier
Evidence-aware model integration tests	5	Memory gate passed at 63 GB per GPU (limit 100 GB); byte-identical checkpoint resume; distributed fault ha

3. Preparatory work on campus resources (no ACCESS charge)

- Built 28 leakage-filtered evidence databases for 9 plant species from ESTs, PacBio, ONT and protein alignments. Each species passed exhaustiveness of at least 0.999, zero leakage and all integrity checks.
- Merged them into one 786 GB evidence database and re-validated the training corpus against it; 37.85 million molecules were consistent with the leakage repair.
- Built a 654-million-row training cache in 455 shards. Every input is bound by cryptographic hash and re-verified at each job start.
- All inputs are staged on DeltaAI, and the first evidence-aware training run is ready to submit.

4. Remaining work and GPU-hour requirement

Measured on DeltaAI with 4 GPUs: about 45 GPU-hours per epoch for the baseline model and about 49.5 for the evidence-aware model. Early stopping (patience 3, maximum 22 epochs) is expected near epoch 18. Our protocol requires the confirmatory baseline seeds 456 and 789 to finish before submission.

Work	Expected	Maximum
Finish baseline seed 123	270	411
Baseline confirmatory seed 456	812	992
Baseline confirmatory seed 789	812	992
Fixed-budget pilot: evidence-aware model and sequence-only counterpart on the same data (4,099 updates each)	196	196
Full training of both models to early stopping	1,584	1,980
Three fixed-budget controls: shuffled evidence, presence-only evidence, reading-frame channels removed	294	294
Evaluation on held-out panels in 9 species and an independent maize test	50	50
Total needed	4,018	4,915
Remaining balance	849	849
Shortfall	3,169	4,066
Contingency (15%) for queue preemption and one re-run	603	737
Supplement requested	**3,800**	

5. Expected outcomes

A pre-registered, leakage-controlled test of whether transcript and protein evidence improves deep-learning gene annotation in plants, with three random seeds for the baseline and control experiments that attribute any gain to evidence content, evidence detail and reading-frame information. Code, frozen protocols and evaluation panels will be released with the manuscript.